

PROFESSIONAL SUMMARY

Technical Designer with experience building gameplay mechanics, VR interaction systems, and rapid prototyping tools across Unity and UE5. Proficient in C#, C++, Blueprints, NavMesh AI, and procedural generation. Experienced in level design, Agile workflows, Git, and playtesting-driven iteration, bridging design and implementation to deliver polished, scalable gameplay experiences.

TECHNICAL SKILLS

Game Engines

Unity (HDRP, URP, XR Toolkit), Unreal Engine 5 (Blueprints)

Programming Languages

C++, C#

Gameplay & Systems

Gameplay Systems, VR Interaction Systems, Procedural Generation, Finite State Machines, AI Behaviour Systems

Tools & Technologies

Git, Unity XR Toolkit, NavMesh, Unity UI Toolkit, Jira, Confluence

3D & Design Tools

Blender, Unreal UMG, Figma

GAME DEVELOPMENT PROJECTS

A Day in My Life — VR Narrative Puzzle Game

Unity | C# | Blender | XR Toolkit | Oculus VR | 2025

- Developed a 3-scene VR puzzle experience in Unity (XR Interaction Toolkit).
- Implemented object grabbing, fragment rotation, teleport locomotion, and climbing mechanics in C#.
- Programmed two puzzle systems (leaf assembly, spiral alignment) using snapping and trigger detection.
- Engineered a Narrative Director system to trigger narration, subtitles, and scene progression.
- Optimized VR scene performance by replacing high-cost 3D grass assets with lightweight textures.
- Evaluated gameplay through peer playtesting (4.6/5 atmosphere, 4.2/5 immersion).

Signal Lost — Scan & Mend System

Unreal Engine 5 | Blueprints | C++ | In Development | 2025–2026

- Designed and built a scan-and-repair mechanic from scratch in UE5 Blueprints for an original sci-fi game.
- Implemented Line Trace by Channel from player camera to detect and validate broken objects in the world.
- Engineered dynamic hologram spawning using Get Forward Vector and world location offsets, with sine-wave hover and rotation via Event Tick.
- Built BP_Mendable_Objects parent class with Construction Script — Broken Mesh, Fixed Mesh, and Material assigned per object directly from the Details panel with zero repeated setup.
- Developed scalable designer-friendly system enabling new broken objects to be configured instantly when dropped into any level.
- Created hologram material from scratch in UE5 Material Editor.

Treasure Sea-ker — 3D Underwater Platformer

Unity | C# | Unity UI Toolkit | GitHub | 2024 | Treasure Sea-ker WebGL Playable Build- <https://jsrgolu.itch.io/treasure-sea-ker>

- Collaborated in a 4-person team to develop a 3-level underwater platformer with AI enemies and inventory systems in Unity.
- Implemented gameplay UI systems including inventory, minimap, and HUD using Unity UI Toolkit.
- Programmed player controls including movement, sprinting, jumping, and weapon switching in C#.
- Developed an inventory system supporting collectible, usable, and equipable items.
- Implemented enemy AI behaviour using a 5-state FSM (Idle, Patrol, Chase, Attack, Dead) with NavMesh.
- Added scalable difficulty modes adjusting enemy health, damage, and spawn counts.
- Improved gameplay through iterative playtesting (functionality score 7.31 → 8.72).

Dual-Control — Custom 2D Game Engine Project

C++ | Visual Studio Code | 2025

- Built a 2D action game from scratch in C++ without a game engine, implementing core gameplay systems.
- Developed systems for character movement, input handling, and collision detection.
- Implemented a dynamic input-switching system allowing real-time toggling between control schemes.
- Focused on memory management and low-level event handling, demonstrating understanding of engine fundamentals.

PROFESSIONAL EXPERIENCE

Technical Support Analyst

Tata Consultancy Services, Mumbai | Client: Westpac Banking Corporation

Jul 2022 – Aug 2024

- Investigated and resolved backend incidents for Westpac's CBF (Customer Business First) banking platform supporting enterprise banking operations.
- Monitored scheduled batch jobs responsible for financial data transfers across production systems.
- Diagnosed and recovered failed data pipelines using SQL scripts and backend troubleshooting.
- Investigated system errors and performance issues through log analysis and debugging.
- Documented incident recovery procedures and operational workflows to support faster troubleshooting and team knowledge sharing.
- Worked in Agile support environments using Jira and Confluence for incident tracking and documentation.

EDUCATION

Queen Mary University of London, Mile End, United Kingdom

(Sep 2024 – Sep 2025)

Msc Computer Games

University of Mumbai- Vishwaniketan's Institute of Technology, Khopoli, Raigad, India

(Aug 2019 – Jun 2022)

BE in Computer Science

(CGPA:7.86/10)

COMMUNITY & ACTIVITIES

- Participated in game jams on itch.io, collaborating on rapid game prototypes.
- Attended AI in Games conferences to explore emerging technologies in game development.
- Member of the Women in Games community.
- Completed training in animation, rigging, and rendering using Blender.
- Active in university arts community, with artwork featured on the college social media page.